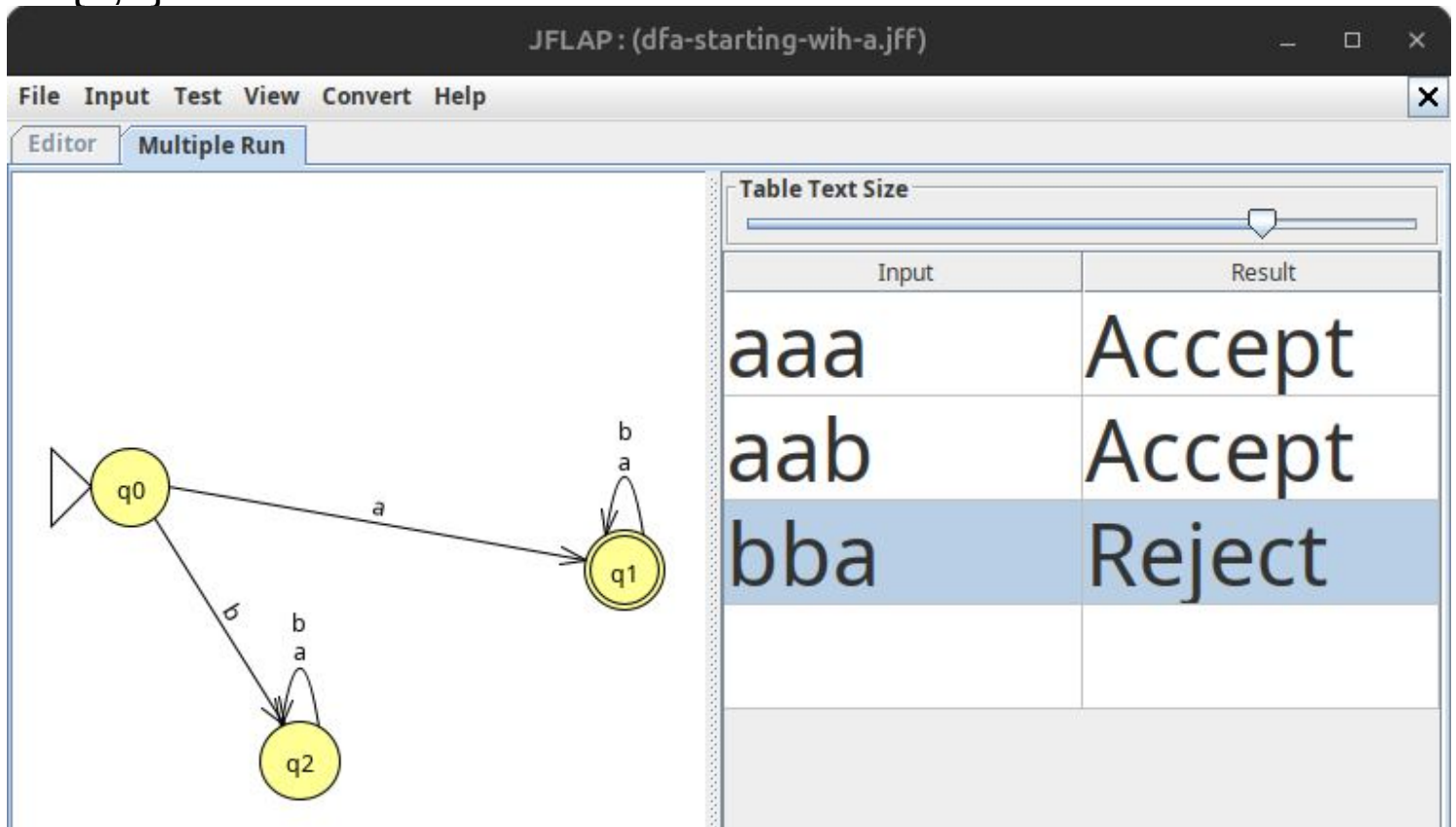
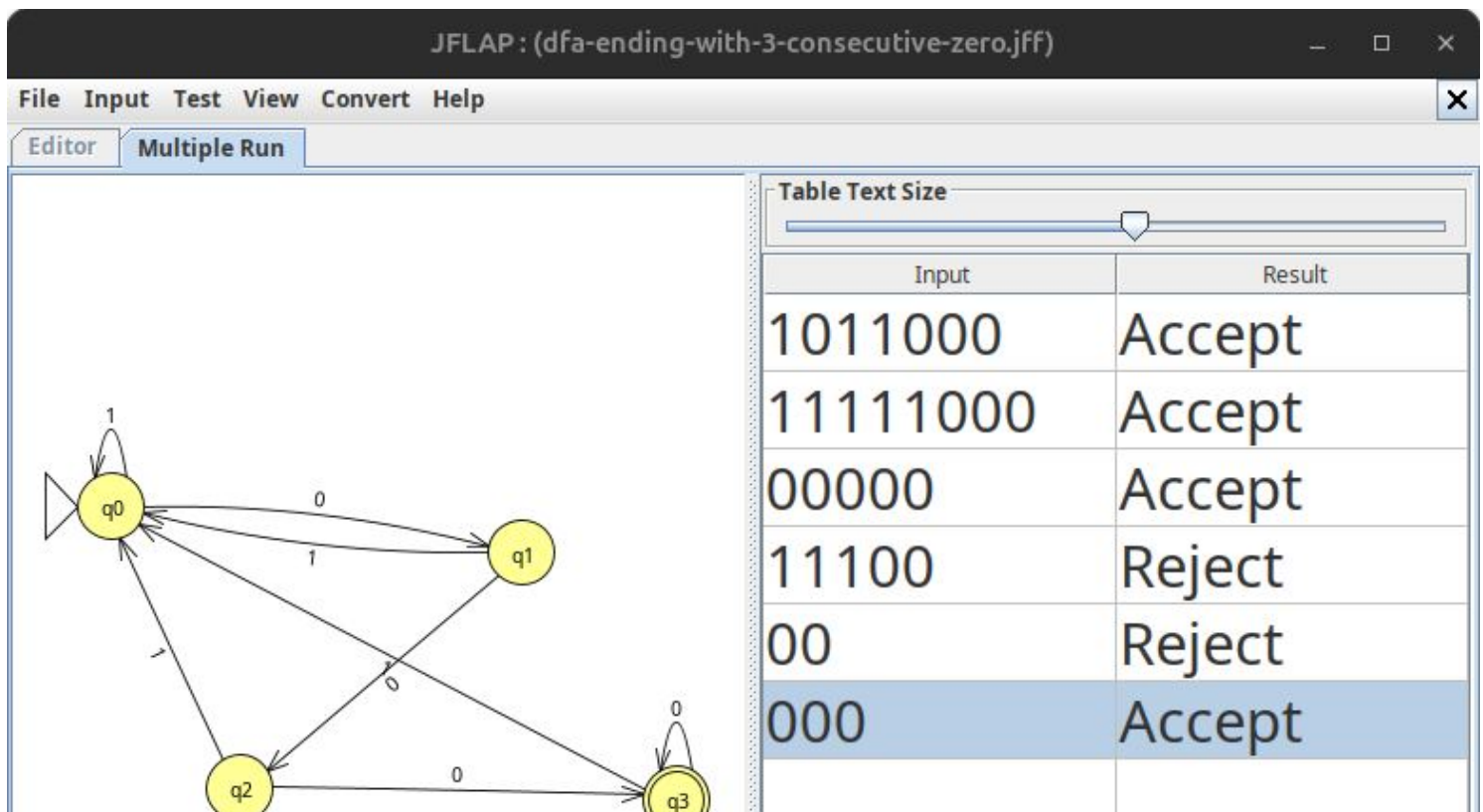


Name: Bishnu Chalise

- DFA set of all strings starting with a, over alphabet = {a,b}



- DFA set of all strings ending with three consecutive zeros over alphabet={0,1}



Name: Bishnu Chalise

- NFA, set of all string containing abab as a substring over alphabet= $\{a,b\}$

Table Text Size

Input	Result
baabab	Accept
baababb	Accept
aabababb	Accept
aaaa	Reject
bbbb	Reject
babaab	Reject

Terminal: rudy@rudy: ~/Documents/4thsem/CSIT_4th

ϵ -NFA, starting or ending with aa or bb over alphabet= $\{a,b\}$

Table Text Size

Input	Result
aab	Reject
abb	Reject
aabb	Accept
abba	Reject

Terminal: rudy@rudy: ~/Documents/4thsem/CSIT_4th_Sem/Theory Of Computa...
rudy@rudy:~/Documents/4thsem/CSIT_4th_Sem/Theory Of Computation/jflp-output\$ nohup java -jar ../software/JFLAP7.1.jar

Traceback windows showing state transitions for inputs: abba, aabb, abb, aab.

- Converting NFA to DFA

Input	Result
ab	Accept
abaa	Accept
bbab	Reject
abba	Accept

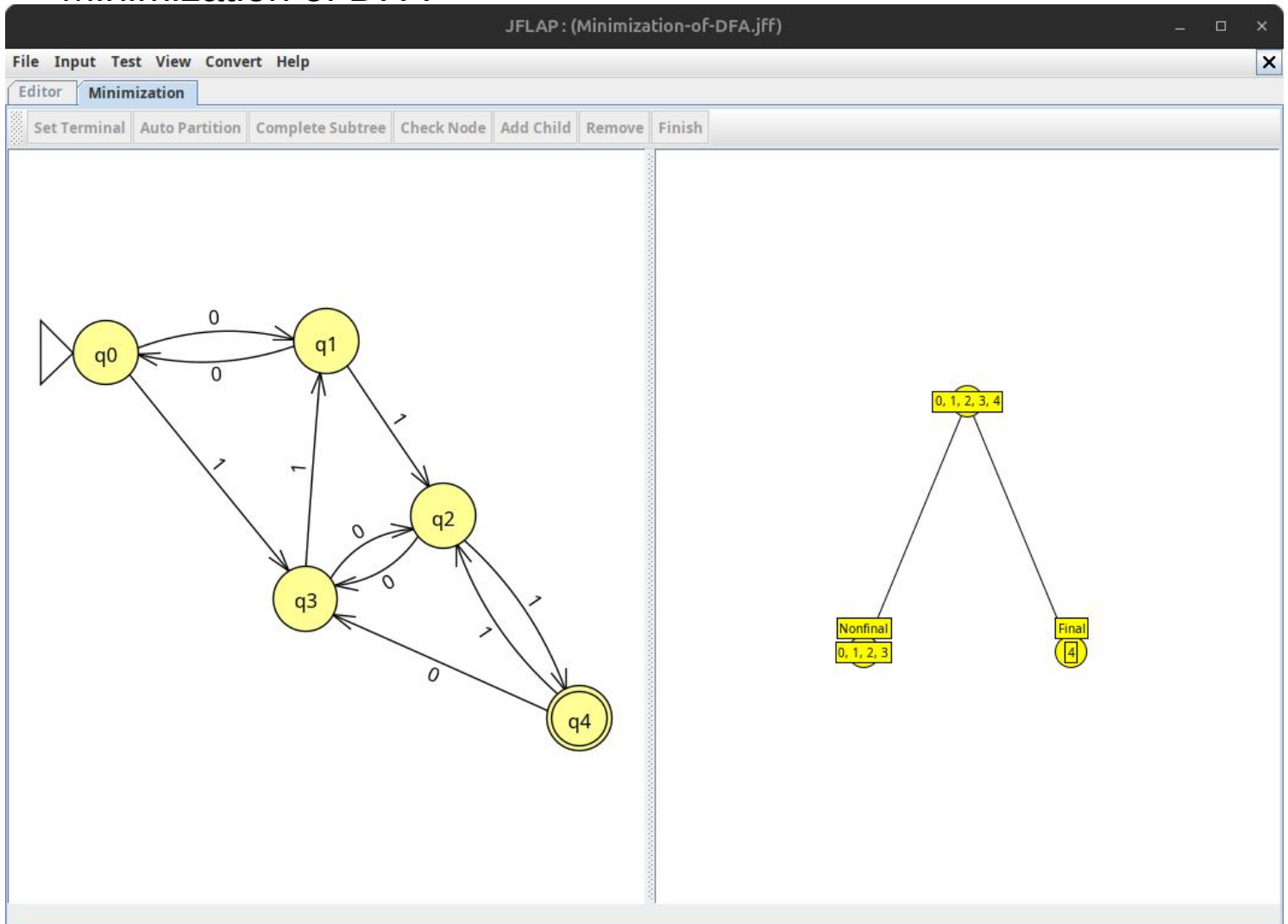
- Conversion, ϵ -NFA to DFA

Accepting configuration found!

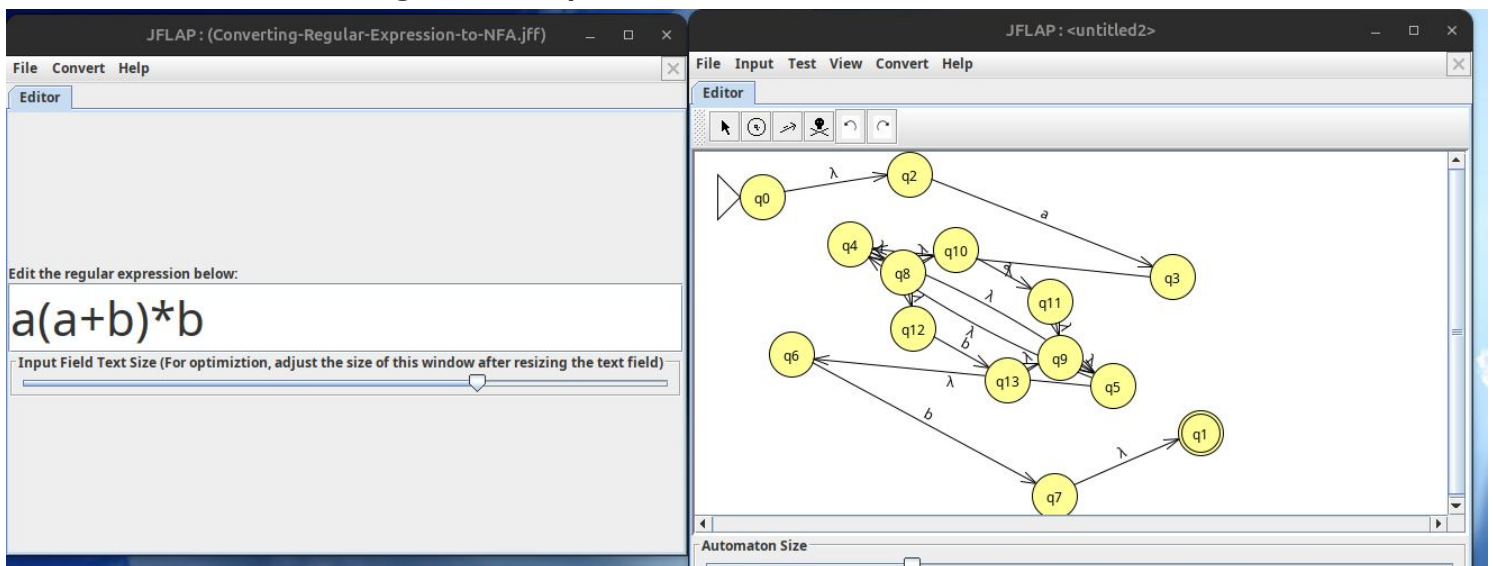
q0	aab
q0	aab
q1	aab
q2	aab

Keep looking I'm done

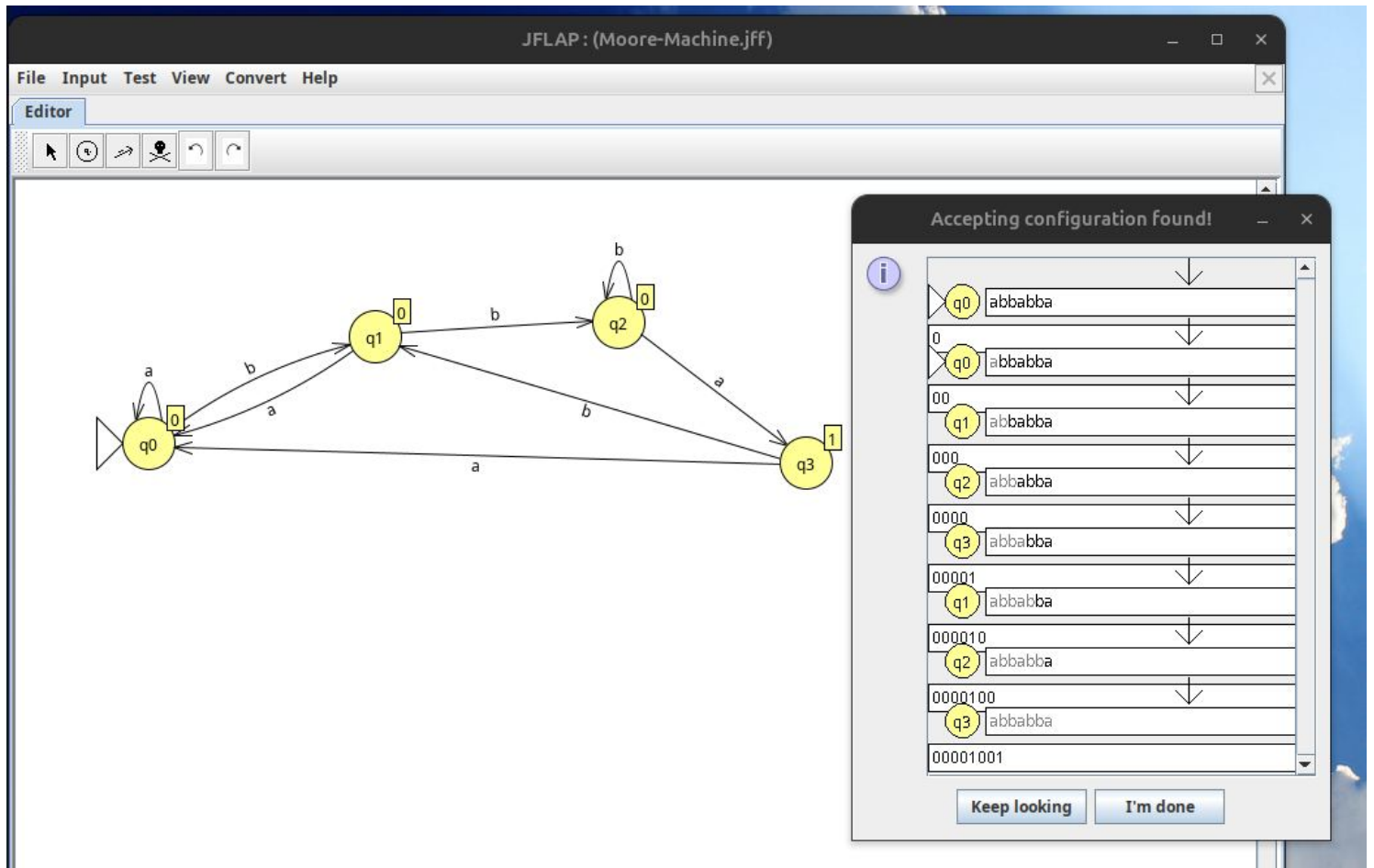
- Minimization of DFA



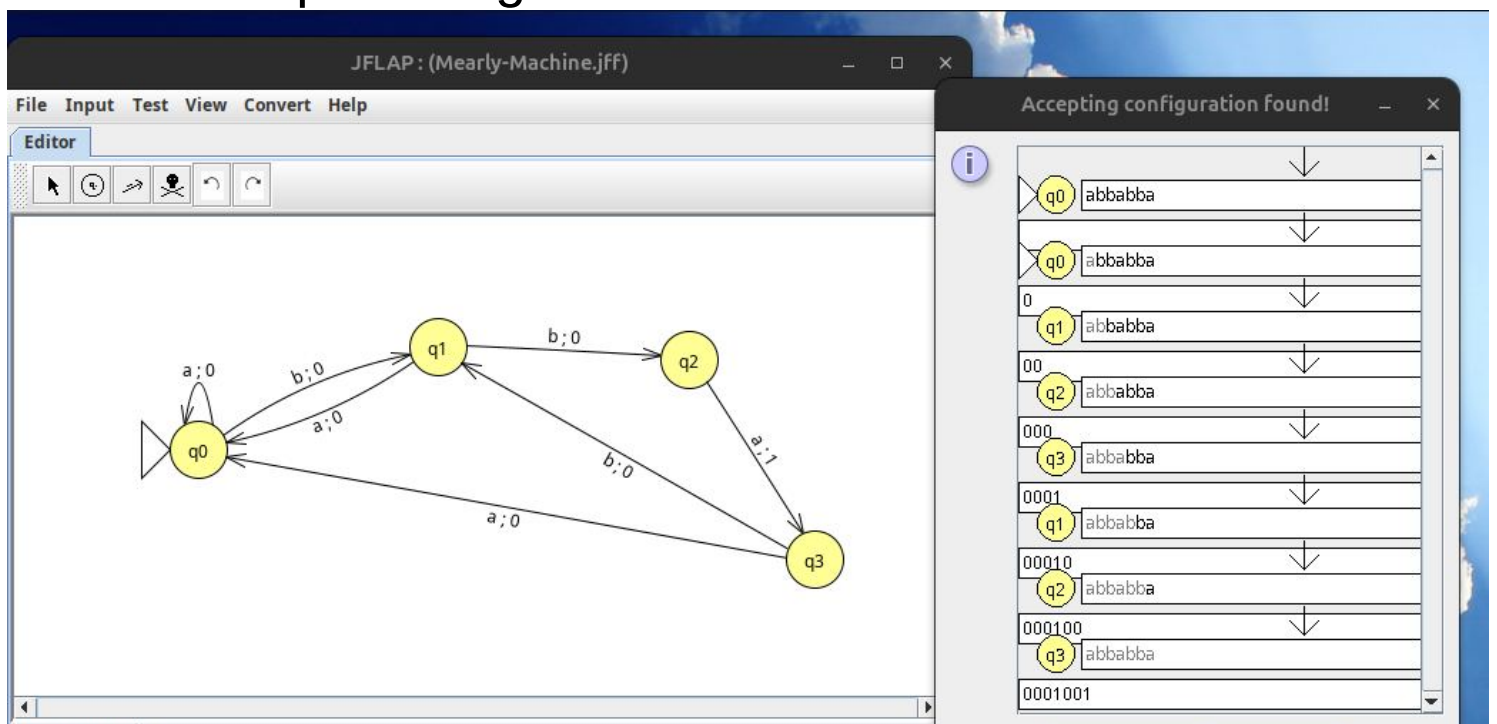
- Conversion, Regular Expression to NFA



- Moore Machine that counts the occurrence of substring 'bba' in input strings.



- Mealy Machine that counts the occurrence of substring 'bba' in input strings.



- Parse Tree – CFG

Input: ababab
String accepted! 9 nodes generated.

Input Field Text Size (For optimization, move one of the window size adjusters around this window after resizing the text field)

Table Text Size

LHS		RHS
S	→	aB
S	→	bA
A	→	aS
A	→	bAA
A	→	a
B	→	bS
B	→	aBB
B	→	b

Derived b from B. Derivations complete.

- Turing Machine ($L = \{a^n b^n \mid n \geq 1\}$)

JFLAP : (Turing-Machine.jff)

File Input Test View Convert Help

Editor Multiple Run

Table Text Size

Input	Result
aabb	Accept
aabb	Accept
abab	Reject
abba	Reject

Traceback

- Push Down Automata ($L = \{a^n b^n \mid n \geq 1\}$)

JFLAP: (push-down-automata.jff)

File Input Test View Convert Help

Editor Multiple Run

```

graph LR
    q0((q0)) -- "a, Z; aZ  
a, a; aa" --> q0
    q0 -- "b, a; λ" --> q1((q1))
    q1 -- "b, a; λ" --> q1
    q1 -- "λ, Z; Z" --> q2(((q2)))
  
```

Table Text Size

Input	Result
abab	Reject
ab	Accept
bbab	Reject
ababab	Reject

Traceback

q0 abab
Z
q0 abab
aZ
q1 abab
Z
q2 abab
Z

Load Inputs Run Inputs Clear Enter Lambda View Trace